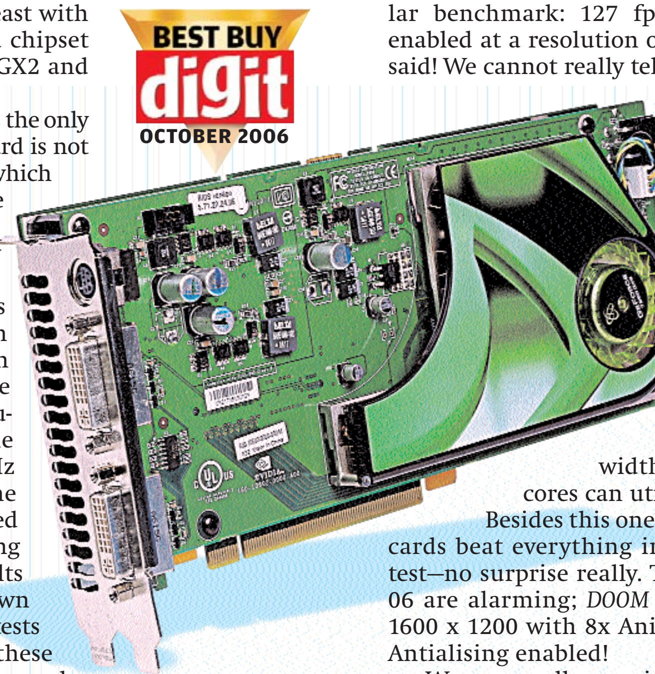
vendors across the board used the stock, double-slot cooler with these cards (NVIDIA reference design). The cooler is designed to throw heat out the rear of the cabinet.

As far as stability goes, we had no problems with the MSI NX7900GTX and the Gigabyte 7950 GX2. The MSI card was actually quite a bit cooler than the other 7900GTX-based cards.

Leadtek's 7900GTX card was plagued with instability issues, and was not able to complete the tests. Another card that was excluded due to stability reasons was the XFX 7950 GX 2 XXX version. It seems the top-end chips are already tweaked to their maximum possible clock speeds, and any attempt by vendors to bump up speeds just causes stability issues—at least with stock coolers. Remember, this is not a chipset fault at all, and both these offerings (GX2 and GTX) are excellent options.

The PX7800GTX 512 from Leadtek was the only 7800GTX 512 chipset in the tests. This card is not to be mistaken with a regular 7800GTX, which ships with 256 MB of memory. The core and memory speeds have been bumped up insanely, and cooling is provided by the 7900GTX type double-slot cooler.

In the red corner, the heavyweights came from Sapphire and ASUS, both based around the powerful ATI Radeon X1900XT architecture. The ASUS card we received was a CrossFire edition. A peculiarity was that both these cards came with their cores running at a 500 MHz default speed (check the table on the previous page for the manufacturer-rated default clocks). The memory was running at 1200 MHz, also well below the defaults (these figures were taken from ATI's own Catalyst utility). We therefore ran our tests at these speeds. One thing is certain: these cards are hot, literally—and this is not a good thing! The cooling solution is a larger version of the older X850XT cards, and throws heat out the rear of the cabinet. These cards easily cross the 80 degree mark while gaming, though ATI does claim that the cores are rated at 110 degrees (so we're well within the danger zone). There is an "XTX" version, which has a 25 MHz bump up in GPU core speeds, and a 50 MHz bump in memory, but we didn't receive this part from any of the ATI vendors.



XFX PV-T71U-ZDD9
Furiously Fast

Performance

The MSI NX7900GTX gave some of the best scores in *F.E.A.R.*, which taxed most cards. The MSI was way ahead of the competition, and bested not only its chipset sibling from XFX, but also the mighty twin core GX2s in this particular benchmark: 127 fps with soft shadows enabled at a resolution of 1280 x 960... enough said! We cannot really tell why the GX2 lost out in the benchmark; it's possible that *F.E.A.R.* scales well according to GPU core and memory speeds. This is compounded by the fact that any CPU and system in general will seriously bottleneck a 7950 GX2-based graphics card, due to the enormous bandwidth that the twin G71 cores can utilise.

Besides this one hiccup, the GX2-based cards beat everything in sight at every other test—no surprise really. The scores in 3D Mark 06 are alarming; *DOOM 3* saw 100+ frames at 1600 x 1200 with 8x Anisotropic Filtering and Antialising enabled!

We were really surprised to see the X1900XT-based cards lagging behind. This may have been due to the core and memory clocks being

JARGON BUSTER

Shaders (Pixel & Vertex): Pixel Shaders (also known as Fragment Shaders) and Vertex Shaders are, very simply, instructions that run on a graphics card. These instructions are executed once for every Pixel or Vertex in a 3D scene. The Pixel Shader is a visual refinement over the Vertex Shader, and owes its existence to the latter. Pixel Shaders operate on a per-pixel basis, unlike Vertex Shaders, which operate with much larger areas (dimension-wise). The Pixel Shader is responsible for that ambient and realistic feel to the surfaces and materials in a 3D scene. The leather texture of a car seat, the metallic look of a gun barrel, the wrinkles on a characters face, all are possible because of Pixel Shaders. Manipulation of light sources as well as the properties of objects in a scene (wood should have that grainy look, metal would possess a certain sheen) is how designers manipulate complex 3D scenes, all using Pixel Shaders.

Bump Mapping: Quite simply, Bump Mapping is a technique used to add detail to a 3D image without increasing the polygon count. In other words, Bump Mapping doesn't interfere with the geometry or positioning of a body in a 3D scene, but rather influences the lighting in such a way so as to apply a texture pattern on an object. Lighting effects (or the lack of light) can be manipulated to create very realistic surface effects. Earlier, EMBM (Environment Mapped Bump Mapping) was used a lot; these days, Displacement Mapping and Parallax Mapping are more used. Games like *F.E.A.R.* and *Oblivion* make heavy use of such techniques.

Filtering: There are three techniques used for filtering. The first technique is called "Bilinear Filtering," which gives the least hit in performance. Very simply, Bilinear Filtering aims to smooth out the overlapping edges in textures that occurs when the image as a whole (or the original pixels individually) are resized.

Trilinear Filtering is a visual improvement over Bilinear Filtering, but the trade-off is reduced frames. However, Trilinear Filtering can be activated on all modern cards without any noticeable loss in frames, and the difference in visual experience is well worth the frame drop.

Anisotropic Filtering (AF) is a relatively newer method of filtering; it is tasked with enhancing the quality of textures on distant objects in a scene. For example, a long road will probably look hazy with AF off. Enabling AF will add depth and realism the distant fringes of the road, replacing the normal textures with much higher-quality samples. AF is very resource-heavy, and most low-end cards will falter with AF set to insane levels. On a high-end system enabling AF will result in some sweet eye-candy.

Anti Aliasing: Removal of unwanted jagged edges that typically plague textures especially when a high resolution signal is displayed at a lower